EEE205 Logic Design Laboratory Project– 9. Lab
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EEE 205 Logic Design Laboratory Manual: Experiment 9

Simple Voting System

Learning Objectives

1. To understand the functionality of 3-to-8 decoders and their role in digital logic circuits.
2. To design a simple voting system using decoders, OR gates, and NOT gates, without the use of memory elements.
3. To analyze input combinations and determine conditions for a tie (TIE LED) and a win (WIN LED).
4. To implement and test the circuit on a breadboard, developing practical skills in wiring and troubleshooting.
5. To visually represent logical conditions using LEDs and evaluate the circuit's performance under all possible input scenarios.

Components Required:

- Breadboard
- Switches
- LED’s
- Trainer Kit
- Connecting Wires
- IC 74HC138 (3-8 Decoder)
- IC 7408 (AND)
- IC 7432 (OR)
- IC 7404 (NOT)

Theoretical Background

This experiment focused on using decoders, OR gates, and NOT gates to design a four-input voting system. The key components are as follows:

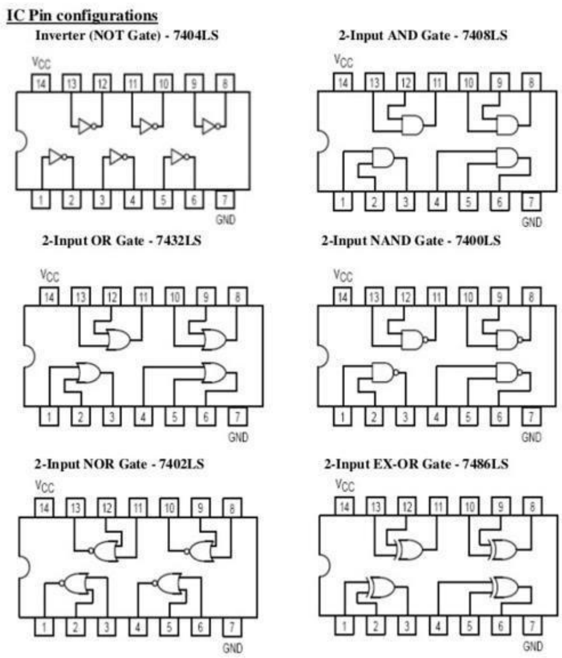
- Decoder (3-to-8): Converts binary inputs into one active output among the eight possibilities. Two decoders were used to handle four inputs and generate signals for WIN and TIE conditions.
- OR Gate: Combines multiple input signals and outputs 1 if any of the inputs is 1. It was used to detect tie scenarios where exactly two participants voted.
- NOT Gate: Inverts the input signal, ensuring proper logic for the decoders and OR gates when necessary.

Voting Logic:

Tie Condition: Exactly two participants voting activates the TIE LED.

Win Condition: Three or more participants voting activates the WIN LED.

Other combinations (e.g., no votes or one vote) result in no LED activation.



Procedure

Circuit Design and Setup:

- The voting system was designed using two 3-to-8 decoders (74HC138). Four inputs (A, B, C, D) represent the participants' votes.
- The outputs of the decoders were connected to an OR gate to determine tie conditions (TIE LED).
- A NOT gate was used where necessary to invert the outputs and ensure proper logic conditions for certain scenarios.
- Two LEDs were implemented to represent the winning (WIN) and tie (TIE) conditions visually.

Implementation:

- The circuit was constructed on a breadboard with switches representing inputs and LEDs showing the output conditions.
- The two decoders were configured to cover all input combinations and provide signals for determining the tie and win conditions.
- OR gates were used to combine specific outputs from the decoders to identify when a tie occurred (two participants voted).
- Proper resistors were used to protect the LEDs, and the connections were thoroughly checked for accuracy.

Testing and Data Collection:

- All possible input combinations (e.g., A=1, B=1, C=0, D=0) were tested.
- Observations were recorded in a truth table format, and the behavior of the WIN and TIE LEDs was verified.
- Scenarios were analyzed to ensure the circuit worked correctly under every condition:
  - **Tie:** If exactly two participants voted.
  - **Win:** If three or more participants voted.

Troubleshooting:

- If the LEDs did not light as expected, the wiring was checked for loose or incorrect connections.
- Adjustments were made to ensure the outputs of the decoders fed correctly into the OR gates.

Truth Table of Simple Voting System

A	B	C	D	WİN	TİE	LOSS
0	0	0	0	0	0	1
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	1	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	1	0
1	0	1	0	0	1	0
1	0	1	1	1	0	0
1	1	0	0	0	1	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	1	0	0

K-Map of winning, tie and lose situation

Win K-MAP

AB\CD	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	0

Boolean Algebra: BCD+ ABD+ ABC + ACD

Tie K-MAP

AB\CD	00	01	11	10
00	0	0	1	0
01	0	1	0	1
11	1	0	0	0
10	0	1	0	1

Boolean Algebra: A’B’CD+A’BC’D+A’BCD’+ABC’D’+AB’C’D+AB’CD’

Lose K-MAP

AB\CD	00	01	11	10
00	1	1	0	1
01	1	0	0	0
11	0	0	0	0
10	1	0	0	0

Boolean Algebra: A’B’D’+A’B’C’+B’C’D’+A’C’D’

Questions:

1. **Explain the primary function of a 3-to-8 decoder. How was the 3-to-8 decoder used in this project, and what logical processes did it perform?**

A 3-to-8 decoder converts a 3-bit binary input into one active output among eight.

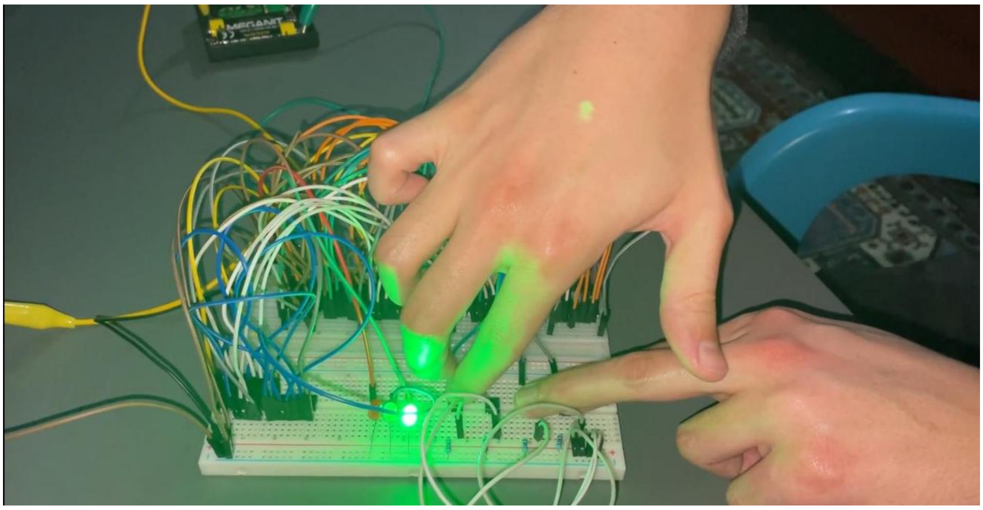
In this project, two decoders processed four inputs (A, B, C, D) to generate signals for "WIN" and "TIE" conditions, simplifying the voting system's design and troubleshooting.

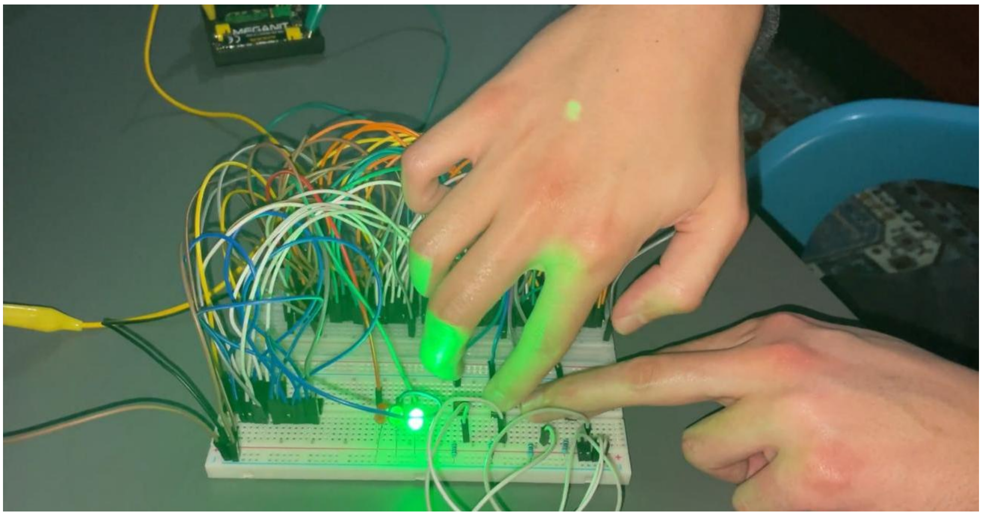
2. **What practical skills are gained because of the experiment?**

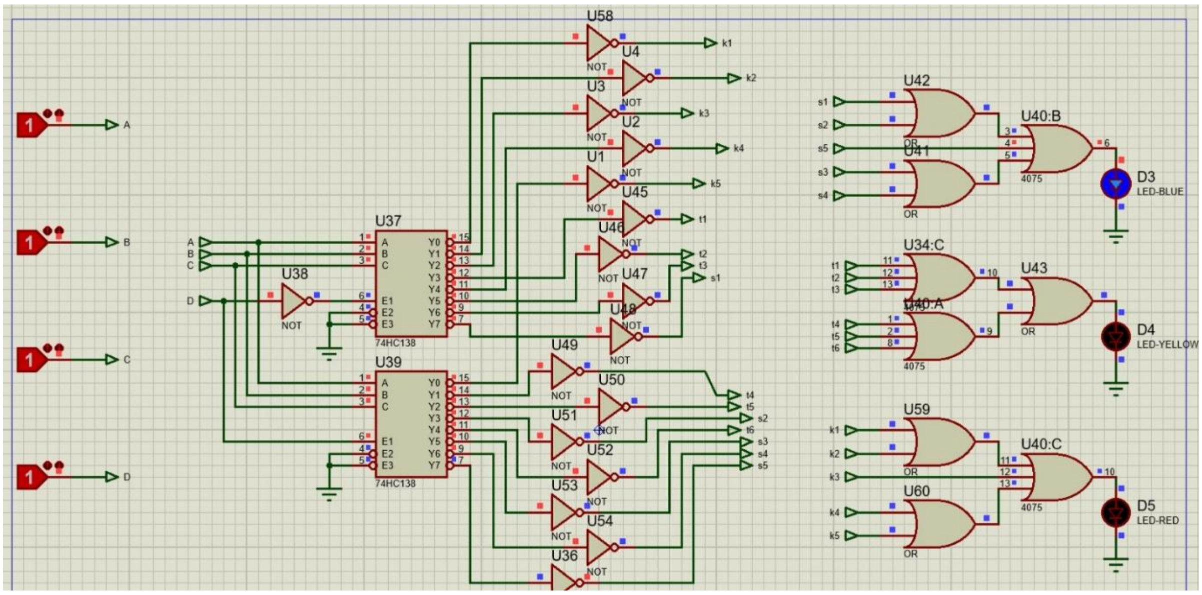
- **Breadboard Usage:** Building circuits on a breadboard and ensuring correct connections.
- **Integration of Circuit Components:** Properly integrating components such as decoders, OR, NOT gates, and LEDs in the circuit.
- **Troubleshooting:** Detecting incorrect connections, testing the circuit, and making necessary corrections.
- **Logical Analysis:** Testing different input combinations and evaluating the logical accuracy of the results.
- **Application of Theory:** Experiencing the functionality of logic gates and decoders in practice.
- **Understanding and Following Circuit Diagrams:** Managing the transition from a diagram to the physical circuit.
- **LED Usage and Protection:** Using LEDs to visually represent output states and protecting them with appropriate resistors.

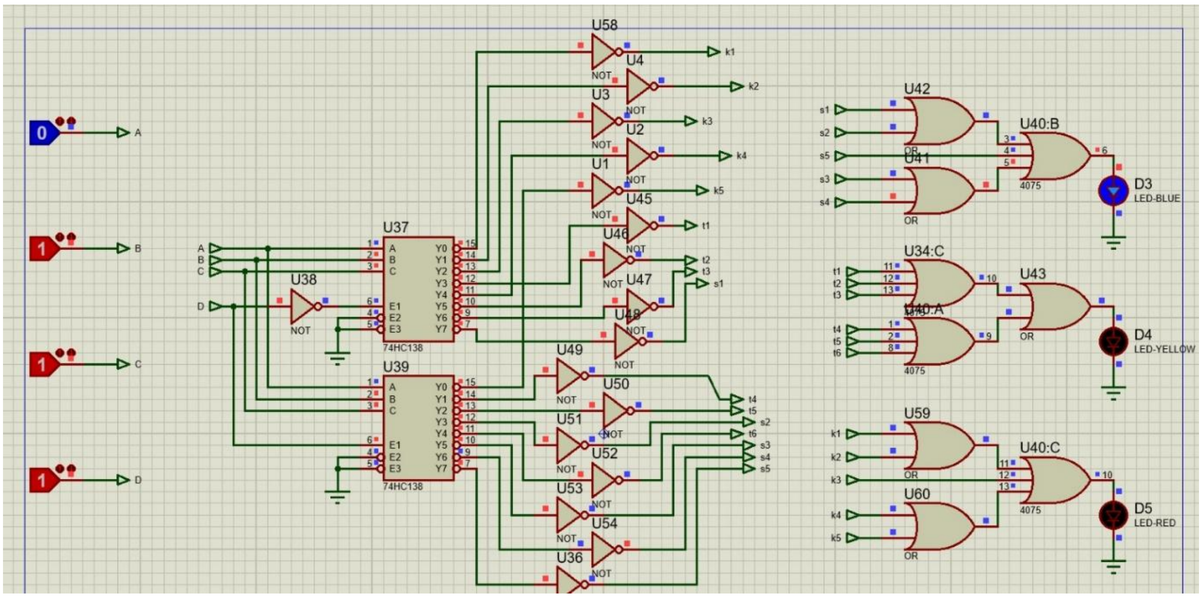
Simulations and Lab Photos

Winning Situations

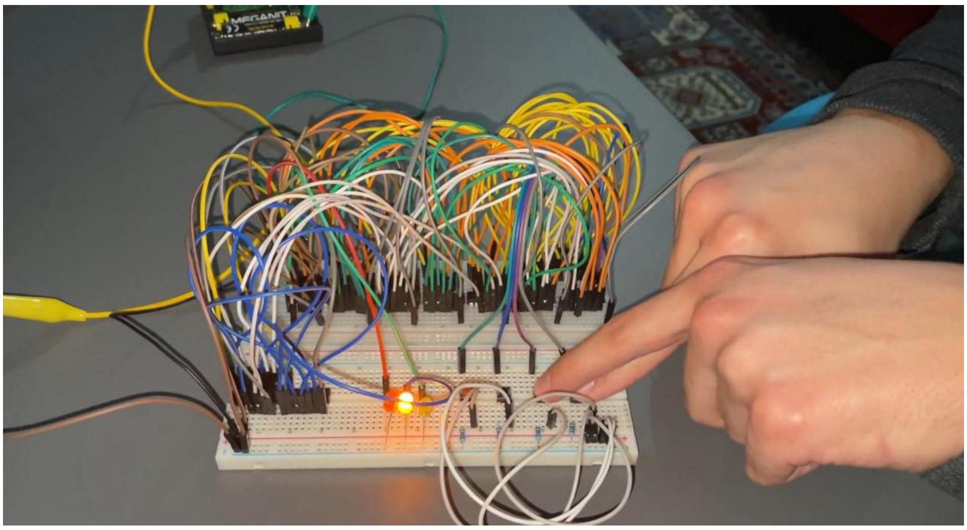


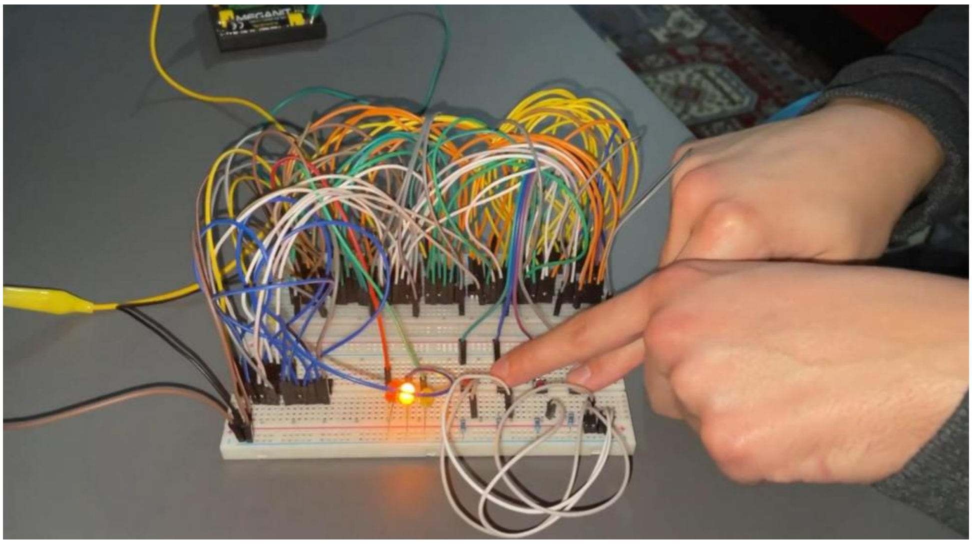


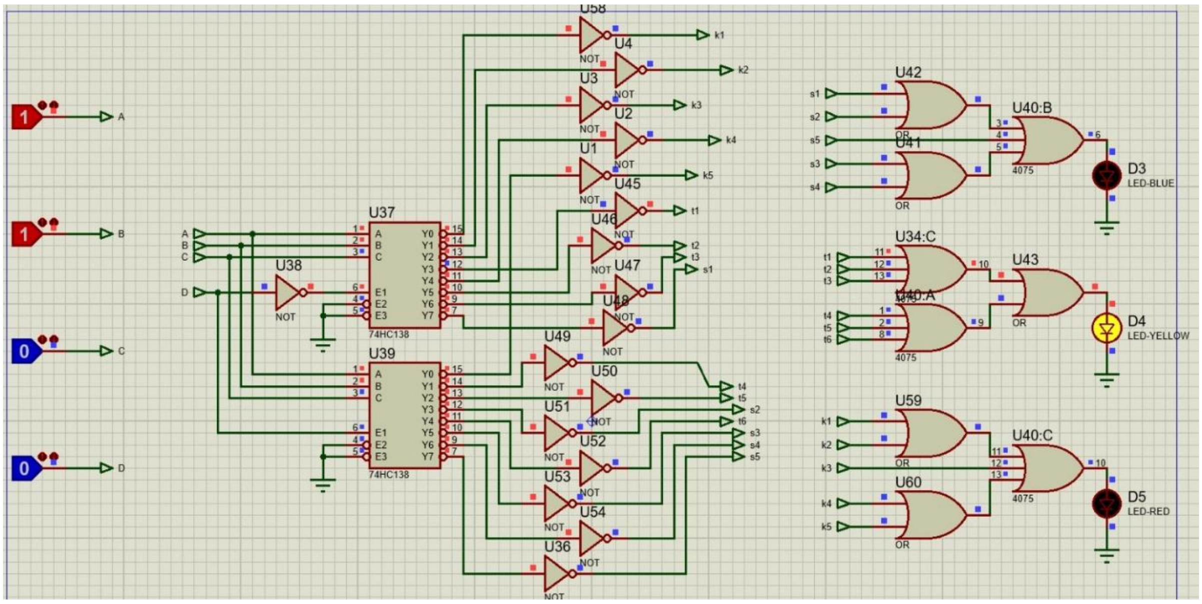


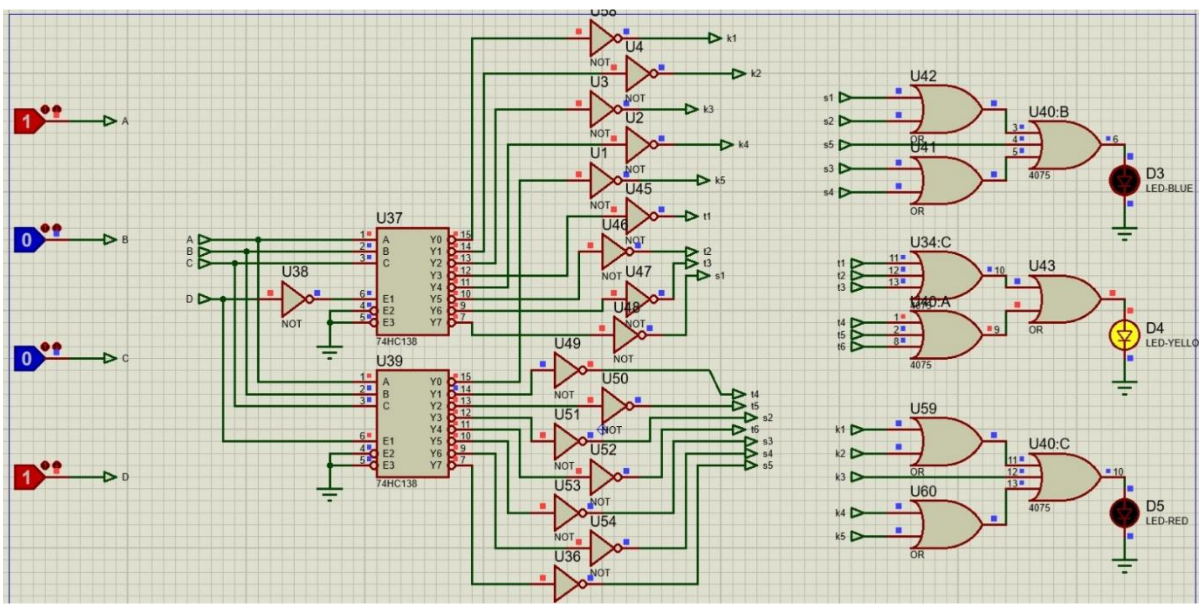


Tie Situations

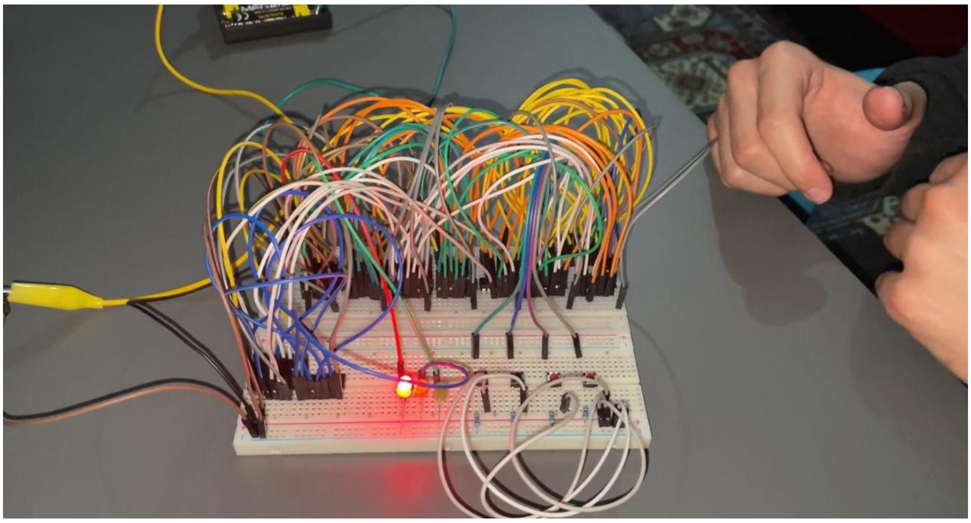


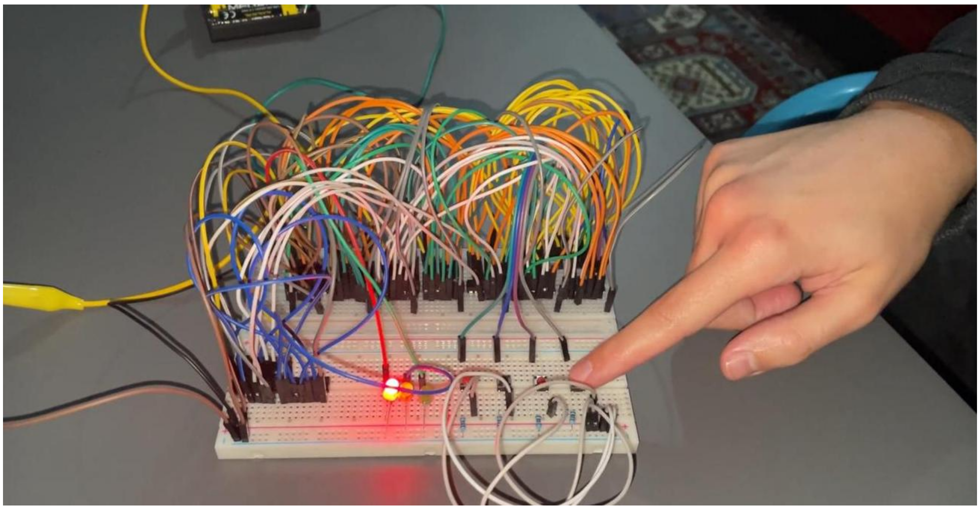


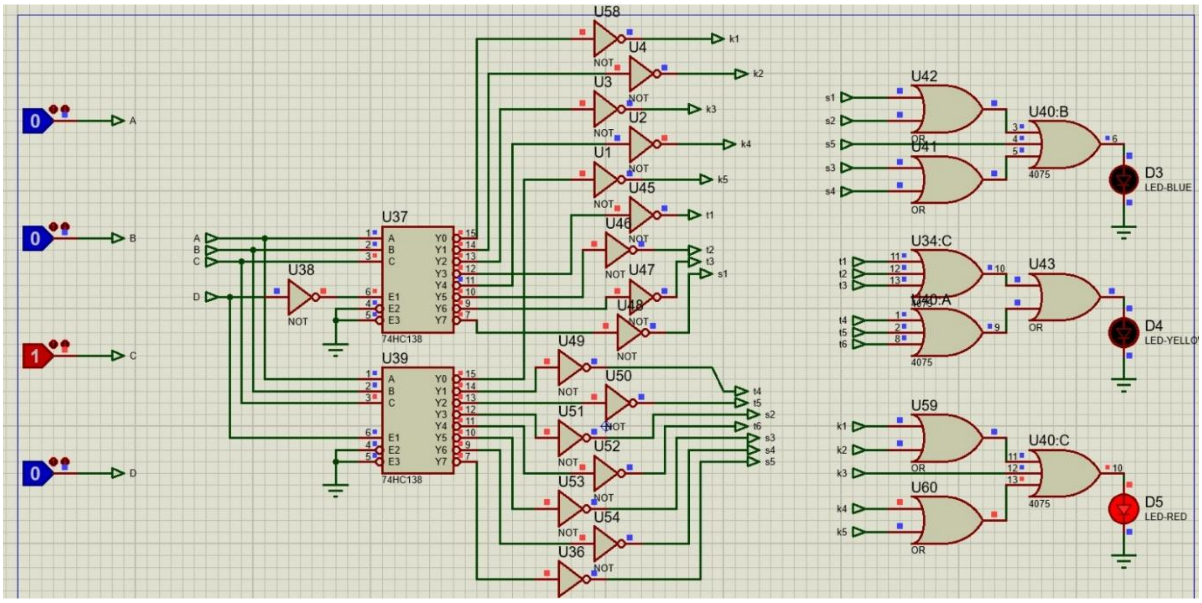


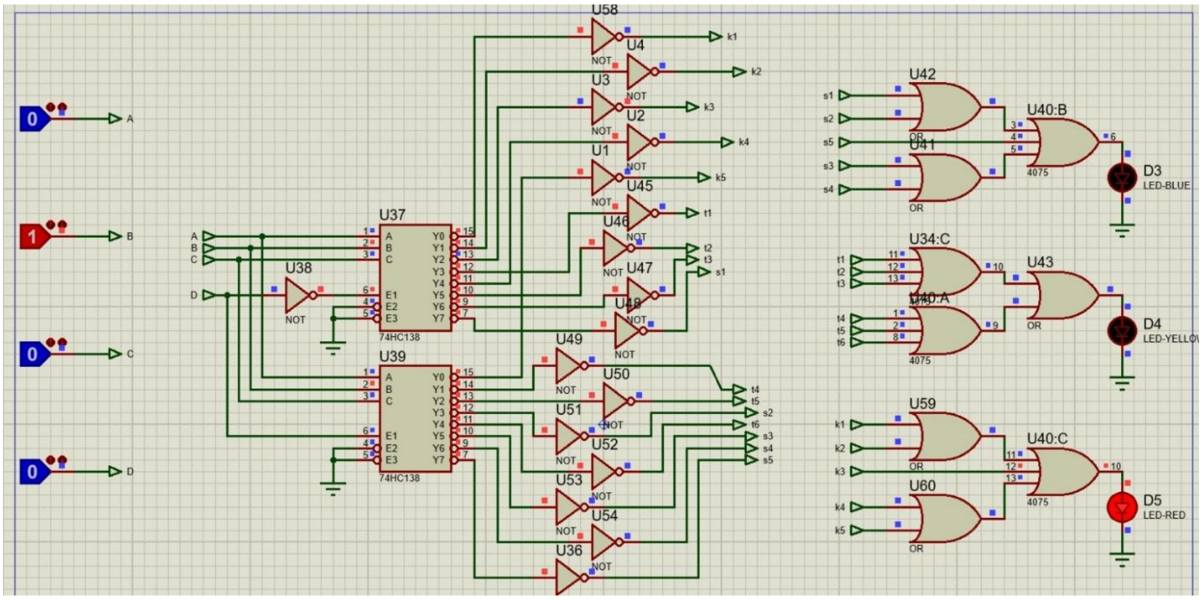


Loosing Situations









Conclusion

This experiment successfully demonstrated the use of digital logic elements, including decoders, OR gates, and NOT gates, to create a simple voting system. The visual representation of WIN and TIE conditions using LEDs reinforced the understanding of logic design principles. Key takeaways include:

1. **Decoder Utilization:** The 3-to-8 decoders effectively mapped input combinations to outputs for logic processing.
2. **Logical Analysis:** The OR and NOT gates ensured accurate processing of voting conditions, highlighting the importance of combining logic gates for complex behavior.
3. **Practical Skills:** Breadboard assembly and circuit troubleshooting provided hands-on experience in digital circuit design.

By the end of the experiment, theoretical concepts such as logic gates, decoders, and signal processing were solidified, and practical application skills were enhanced.